WASP-52b

R-0433

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-52_171123` · created 2026-07-19T11:57:46+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458080.6429 +/- 0.0016 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.165 +/- 0.011
Transit depth (Rp/Rs)^2	2.72 +/- 0.35 [%]
Orbital Inclination (inc)	84.92 +/- 0.71
Airmass coefficient 1 (a1)	1.1496 +/- 0.0092
Airmass coefficient 2 (a2)	-0.1121 +/- 0.0079
Scatter in the residuals of the lightcurve fit is	0.8 %
Best Comparison Star	None
Optimal Aperture	3.72
Optimal Annulus	6.72
Transit Duration (day)	0.0728 +/- 0.0049

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.165 ± 0.011 vs 0.1646 (deviation 0.04σ)
Duration fit vs published	0.0728 d vs 0.0758 d (deviation 0.62σ)
Mid-time O–C	-4.0 min
Depth SNR	15.0
Residual scatter	0.8 %

Light curve

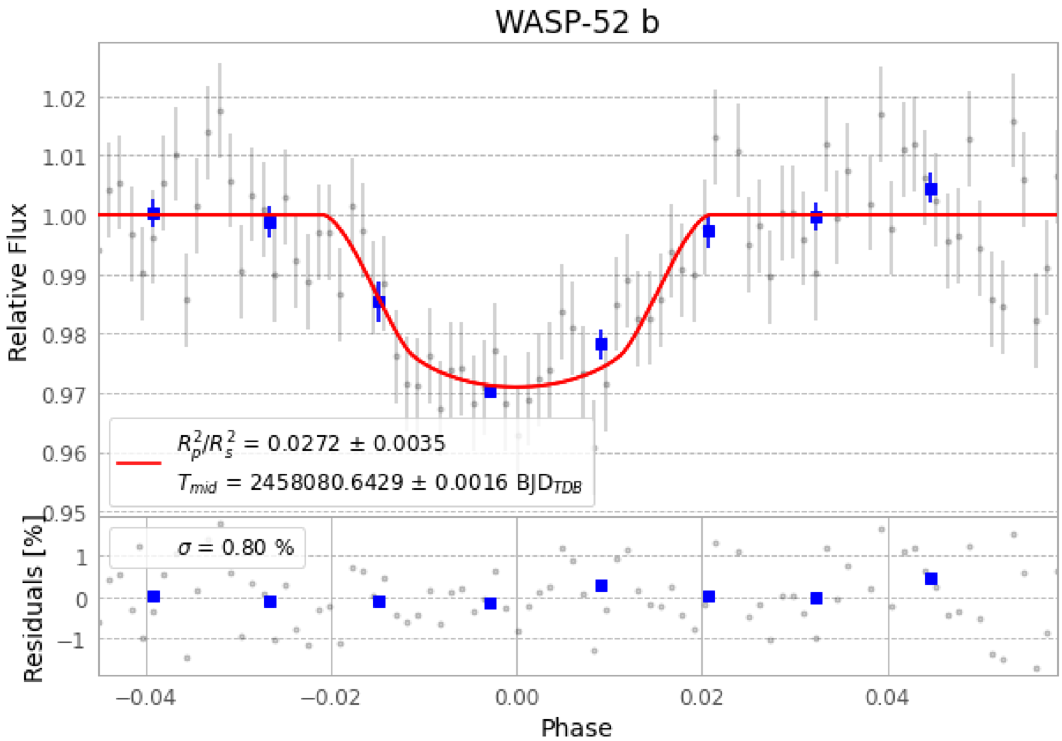

Figure 1 — FinalLightCurve_WASP-52 b_2017-11-22.png (SHA-256 808a756faa46cde4…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [305, 236], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 5341deb97e2bf118…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80`

Data provenance

88 FITS frames (58 MB), `WASP-52171123012712.FITS` → `WASP-52171123054812.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-171123.log (93b0fbe45be0…), FinalLightCurve_WASP-52 b_2017-11-22.png (808a756faa46…), FinalLightCurve_WASP-52 b_2017-11-22.csv (ee82c99f1981…)

Compute resources

Wall time 115.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)